

Fig. 3: Example of a distorted document.

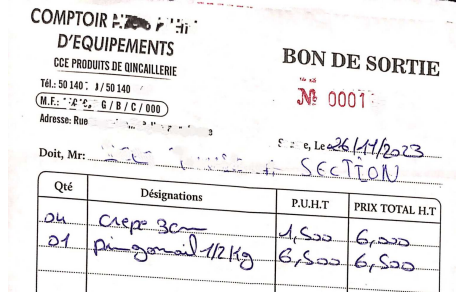


Fig. 4: Example of a skewed document.

2) Data augmentation and preparation:

In order to improve the ability of the DL models to generalize, data augmentation and preparation techniques are required due to the limited size of collected dataset. Given that the adopted workflow is divided into two main tasks, namely keyword detection and object detection, each task is subject to specific and adapted augmentation techniques.

For the keyword detection task, the challenges encountered by a DL model include the presence of unclear characters and image distortions such as rotation, shear, or scale transformations. Therefore, we choose to augment the dataset using the following techniques:

- Median blur and color jitter: improves clarity by reducing noise and simulating various lighting conditions, particularly in documents with complex backgrounds.
- Padding: prevents critical text from being cut off during data preparation, with a probability set to 1.
- Rotation: is limited to 5 degrees using inter-cubic interpolation to minimize data loss. In order to preserve information and standardize image dimensions, an additional step ensures no cropping occurs closer than 20 pixels from any bounding box after rotation.
- Resize: is performed using bi-linear sampling to maintain consistent image dimensions, supporting uniform model performance.

For the stamp and signature detection task, our goal is to present these objects under varying conditions. The following outlines the augmentation techniques we apply:

- Rotation: is performed with a maximum of 10 degrees to simulate the common skew in scanned documents. The rotation transformation is applied with a probability of 0.5 to ensure data diversity.

- Color jitter: is performed with a probability of 0.4 to account for color variation and lighting conditions, which enhances the model's resilience in detecting signatures and stamps across different documents.
- Multiplicative noise: is added on a per-channel basis with a multiplier ranging from 0.5 to 1.0, applied with a probability of 0.3. This noise simulates digital scanning artifacts and helps train the model to be less sensitive to such variations.

Figure 5 illustrates the distribution of the final dataset after data augmentation. It is important to note that the class distribution remains unchanged following the augmentation process, as the applied operations only affect the number of instances per class.

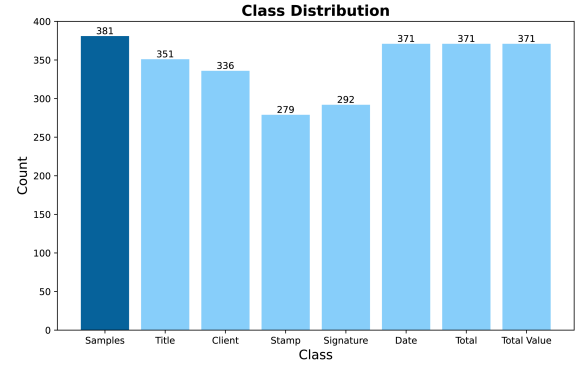


Fig. 5: Final dataset class distribution.

B. Methods

This subsection outlines the deep learning models we plan to train and evaluate for both keyword detection and stamp and signature detection tasks.

1) Keyword detection:

We perform keyword detection to identify significant parts of the document (e.g., customer name, document type, date, etc.). To achieve this, we use (i) an OCR step followed by (ii) a document layout analysis step. This approach is necessary for attention-based models that incorporate embedded OCR to enhance document understanding. Below, we present the selected methods for both steps and provide a comparison to determine the most effective combination of OCR and document layout analysis methods for keyword detection.

An OCR step for character extraction from input images, utilizing robust modules, is essential for the document layout analysis task. Choosing the appropriate OCR model requires comparing the most relevant OCR tools. The most relevant OCR models in the literature are tested on the FUNSD dataset [14], which stands for Form Understanding in Noisy Scanned Documents. To evaluate the similarity between the extracted and the ground truth text in the dataset, we consider two metrics from the original papers: accuracy and similarity based on *Levenshtein* distance [19], which is particularly suited for measuring the distance between characters in a word.